

RESEARCH ARTICLE

Marine zooplankton contribute differentially to community synchrony across a 37-year time series

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Abstract

1. Community synchrony—the tendency for species within a community to fluctuate similarly to other species—is tied to environmental variability through the responses of species in the community to environmental drivers. If species respond similarly to the environment, environmental variability will increase synchrony, as community synchrony is a composite property of species-level dynamics.
2. We used a 37-year time series of marine zooplankton to explore temporal patterns in community synchrony, the relationship between environmental variability and community synchrony, and the relationship between taxa-level contributions to community synchrony and population dynamics.
3. Environmental variability was unrelated to annual community synchrony estimates, taxon density was related to taxon-level contributions to community synchrony, and taxon-level contributions to community synchrony have become less correlated over time (i.e. taxa are not maintaining their relative rank in synchrony contributions year to year).
4. Numerically dominant taxa contributed more to community synchrony, but not because of their numerical dominance. Instead, taxon-scale temporal variability was a strong indicator of contribution to synchrony, suggesting more stable taxa contribute more to overall community synchrony.
5. Collectively, this highlights the inherent link between community synchrony and the population processes that in aggregate constrain resulting community dynamics, while finding no support that more variable environments will alter community synchrony.

KEYWORDS

community synchrony, environmental variation, estuaries, marine zooplankton, population dynamics

1 | INTRODUCTION

Species in ecological communities fluctuate with each other either positively (synchrony) or negatively (compensation) over time (Gouhier &

Guichard, 2014) as a function of species interactions and environmental tolerances, while also being subject to demographic and environmental stochasticity. The resulting covariation in species densities, along with the overall community size, is a key element to estimating community stability

(Loreau & de Mazancourt, 2008; Valencia et al., 2020). Synchronous or compensatory fluctuations among interacting species can be driven by a number of different factors (Loreau & de Mazancourt, 2008; Shoemaker et al., 2022). For instance, strong competition between species would be likely to lead to compensatory dynamics, in which the reduction in density of one species would release the other species from competitive pressure, allowing the second species to increase in density (Gonzalez & Loreau, 2009). While competition is important, the role of differential (or shared) responses of species to environmental drivers is another main determinant of synchrony (or compensatory dynamics). Notably, the temporal variation in environmental conditions may cause ecological communities to reorganize, potentially causing large shifts in relative abundance through time (Scranton & Vasseur, 2016; Tuljapurkar, 2013; Vasseur & Fox, 2007). Exploring how communities respond to competitive and environmental pressures is important to understanding how communities may change with both changes in mean environmental conditions as well as increasingly variable environments (Franklin et al., 2016; Ives, 1995).

Community synchrony is often estimated using all available density time-series data on a given community (Valencia et al., 2020; Vasseur et al., 2014), but synchrony may also change through time (i.e. be higher or lower in certain periods) and/or be timescale-specific (i.e. be found at annual timescales but not at decadal timescales; (Shoemaker et al., 2022; Vasseur et al., 2014)). In spatial population dynamics, increasing synchrony could be a signature of reduced meta-population persistence (Liebhold et al., 2004). Similarly, for community dynamics, increasing synchrony would cause large fluctuations in overall community size, with clear potential to alter the strength of competition, community composition and important ecological interactions such as pollination or parasitism (Lázaro et al., 2022). Given suitable time series data, we might expect that time periods with more variable environments would also tend to have lower community synchrony, under the assumption that species respond uniquely to the environment, or if they track environmental changes at different timescales (Gouhier & Guichard, 2014; Tsang et al., 2023). That is, species will likely differ in the strength and direction they respond to a given environmental condition, such that environmental variation will only exacerbate population dynamic differences between the species by visiting regions of that environmental space where the species may differ in their species–environment relationships. Essentially, in more variable environments, we would expect the correlation between species demographic rates to be weaker, leading to asynchronous dynamics, whereas species responses to a constant environment would correspond to a single point in their species–environment relationship. If species had similar responses to the environment, we might expect environmental variability to either not influence community synchrony at all, or even increase community synchrony, depending on the relative roles of demographic stochasticity and species–environment relationships in driving community synchrony. Finally, an interesting possibility exists in that species may respond identically to the environment, but at different timescales due to generation time differences or other demographic factors, which could still lead to more asynchronous dynamics even in the case of equivalent species–environment relationships.

Apart from species responses to environmental drivers, another important consideration is the relative dominance of species in a community. That is, we might expect species which are more dominant in a community to be less variable through time, as the impact of demographic stochasticity on population dynamics may be reduced (Anderson et al., 1982). The ability to parse out an individual species' contribution to community synchrony would allow a clear test of this idea, with potential to lead to a more general understanding of drivers of community synchrony and potentially to inform how we measure community synchrony (e.g. should we standardize species abundances in a time series to weigh species equally?). Further, species might be expected to have relatively consistent contributions to community synchrony over time, barring large-scale changes in environmental conditions between time periods or for communities with high turnover. The consistency of species contributions to community synchrony could be a signature of overall community stability, suggesting that temporal changes in the relative contributions of different species to overall community synchrony may be important to consider.

Here, we leverage a 37-year time series of marine zooplankton community dynamics sampled every 2 weeks to explore how community synchrony changes through time, the role of environmental variability on community synchrony and how taxa-scale contributions to community synchrony are related to average abundance and time. We expected that environmental variability would result in less synchronous communities, contributions to synchrony would be driven by the numerical dominance of taxa within the community and that the contributions of taxa to synchrony would be conserved over time (i.e. year to year variation in taxa contribution to synchrony would be stable, and the rank order of taxa by their contributions to synchrony would be similar over time). We found that environmental variability was unrelated to community synchrony, taxa-scale abundance was positively related to contributions to community synchrony, and these contributions to community synchrony have become less correlated over time (i.e. less consistent rank order of contributions in more recent years). By standardizing taxa abundance, we were able to parse out the effects of numerical dominance from temporal variability in population dynamics, finding that less temporally variable taxa contributed most strongly to community synchrony. Taken together, this suggests that environmental variability is potentially unrelated to community synchrony in this system, that numerically dominant and/or temporally stable taxa contribute positively to community synchrony and that taxa-scale contributions to community synchrony are becoming less related over time.

2 | METHODS

2.1 | Marine zooplankton sampling

Zooplankton samples were collected from North Inlet Estuary (NIE), near Georgetown, South Carolina, within the North Inlet-Winyah Bay National Estuarine Research Reserve (Figure 1). NIE is a shallow, polyhaline estuary that is surrounded by saltmarsh and

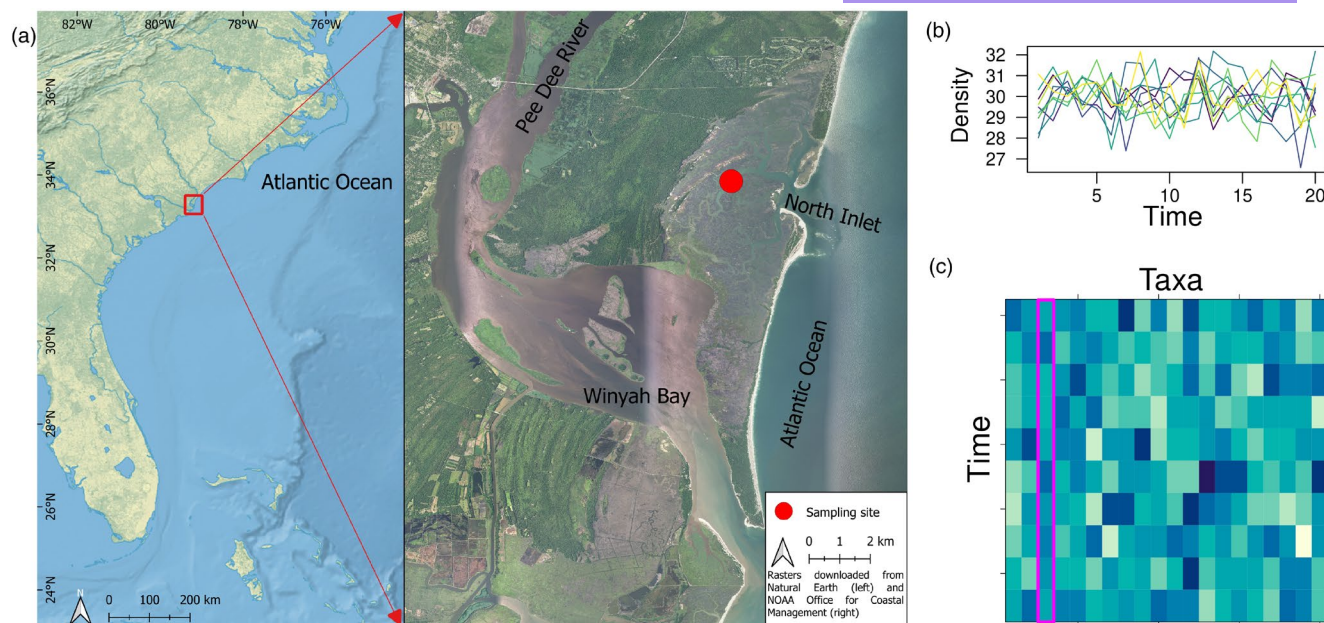


FIGURE 1 Zooplankton communities were sampled from North Inlet Estuary, South Carolina, every 2 weeks for 37 years (sampling site is red point in a). Using these time series data at taxon-level resolution (b), we estimated community synchrony, capturing how species fluctuate relative to one another. By randomizing individual taxa with respect to time (c, with purple box isolating a single taxa over time), we were able to estimate the contribution of each taxa to overall community synchrony, as well as explore how this contribution has changed over time.

has a short residence time (average amount of time a volume of water stays in the estuary) of approximately 15 h (Kjerfve, 1986). It is an ideal system for studying long-term trends, as around 98% of its watershed is undeveloped land (Allen et al., 2014), and thus, changes in the estuary can more readily be attributed to regional, instead of local, environmental changes. Zooplankton samples were collected twice each lunar cycle (approximately every 2 weeks) by towing a round zooplankton net that had a 30 cm diameter opening and was fitted with 153 μ m mesh. Two replicate, 3-min tows were taken simultaneously 2.5 h before low tide between 08:00 AM and 03:00 PM at the same location mid-estuary (33.334, -79.186). The volume of water sampled by each net was calculated by multiplying the area of the mouth of the net (0.071 m²) by the distance travelled through the water as measured by a General Oceanics Flowmeter (Model 2030R) fitted in the mouth of each net. Samples were immediately preserved in 4% borax-buffered formaldehyde and stained with Rose Bengal to aid in separation of animal from plant material.

2.2 | Marine zooplankton identification

Identification and quantification of zooplankton taxa were completed under a dissecting microscope by taxonomic experts. One subsample (1%) of each of the two replicate tows was counted, and the counts from each tow were converted to abundances (individuals per cubic meter). Abundances from the two replicates were averaged together for the purposes of this study. Zooplankton were identified to broad taxonomic/life stage categories, as species-level identifications

were too time consuming to complete for the 1830 samples. Within each of these taxonomic/life stage categories, diversity is relatively low and overlap in ecological roles is high (Allen et al., 2008; Mallick et al., 2024; Stone & Stone, 2023), and each broad taxon can generally be treated as a single functional group. Taxonomic groups identified were appendicularians (Phylum: Chordata, Class: Appendicularia), barnacle nauplii (Phylum: Arthropoda, Subclass: Cirripedia), barnacle cyprids, bivalve larvae (Phylum: Mollusca, Class: Bivalvia), brachyuran crab zoea (Phylum: Arthropoda, Class: Malacostraca, Order: Decapoda, Infraorder: Brachyura), bryozoan larvae (Phylum: Bryozoa), cladocera (Phylum: Arthropoda, Class: Branchiopoda, Superorder: Diplostraca), copepods (Phylum: Arthropoda, Class: Copepoda), doliolids (Phylum: Chordata, Class: Thaliacea, Order: Doliolida), echinoderm bipinnaria (Phylum: Echinodermata), echinoderm plutei, gastropod larvae (Phylum: Mollusca, Class: Gastropoda), hydrozoan medusae (Phylum: Cnidaria, Class: Hydrozoa), ostracods (Phylum: Arthropoda, Class: Ostracoda) and polychaetes (Phylum: Annelida, Class: Polychaeta).

2.3 | Community synchrony

Community synchrony was defined using the formulation of Loreau and de Mazancourt (2008), which has strong conceptual foundations and clear interpretability. It is bound between 0 and 1, where values closer to 1 indicate more synchronous communities, and values closer to 0 indicate randomly fluctuating communities. Note that anti-synchronous communities—those where species fluctuate in opposition to one another—can only be truly estimated for two-species communities, as the addition of a third species will inherently reduce

the anti-synchrony. This is one reason why mean pairwise correlations among community members tend towards 0, as they do capture anti-synchrony, but this measure is quite sensitive to species richness (Loreau & de Mazancourt, 2008). On the other hand, the ϕ measure (Equation 1) is not as sensitive to the number of species in the community as other methods to estimate community synchrony, though it is incapable of capturing anti-synchrony. The resulting community synchrony measure ϕ , is defined as

$$\phi = \frac{\sigma_x^2}{\left(\sum_{i=1}^n \sigma_i\right)^2} \quad (1)$$

This estimate uses the temporal variance of the entire community (σ_x^2) and the standard deviation of each individual species ($\sigma_{x,i}$) as the key elements of community synchrony (Equation 1). To assess significance of community synchrony, we used the `community.sync` function in the R package `synchrony` (Gouhier & Guichard, 2014), which randomizes species abundances over time independently 999 times and estimates z-scores comparing the observed synchrony to the distribution of randomized community synchrony values. z-scores were computed as $\frac{x-\mu}{\sigma}$, where x is the true synchrony, μ is the mean randomized synchrony and σ is the standard deviation of the randomized synchrony.

2.4 | Linking environmental variability to community synchrony

Temperature and salinity variability were defined as temporal variation (standard deviation) in environmental conditions for each year, where each year was sampled either 24 or 25 times. The use of standard deviation does assume that there is no directional mean shift over time in temperature or salinity that might influence standard deviation. We explore this in the supplement by using a proportional variability measure (Heath & Borowski, 2013), finding a very strong relationship between the two measures of variability (Figure S3). At each zooplankton sampling event, salinity and temperature were measured at the surface using a Beckman RS5 conductivity meter (prior to September 1993) or a YSI EXO sonde (after September 1993). We related these values of environmental variability to community synchrony, with the underlying idea that more variable environments should promote asynchrony. Temperature variability and salinity variability were related to community synchrony using linear models. All analyses were performed in R 4.4.2 (R Core Team, 2024). We hypothesized that environmental variability would be negatively related to community synchrony, under the assumption that marine zooplankton taxa have slightly different responses to environmental variation (i.e. response diversity; Ross and Sasaki (2024)). On the other hand, if zooplankton responded similarly to environmental variation and had similar generation times or life histories, variable environments could actually produce more synchronous communities. Given the diversity of life-history strategies and environmental tolerances of these taxa (Allen et al., 2008; Johnson et al., 2012), we expected that environmental variability would be negatively related

to community synchrony. Further, previous work has demonstrated differential demographic responses of marine zooplankton to environmental drivers such as temperature (Dvoretzky & Dvoretzky, 2025; Zervoudaki et al., 2024).

2.5 | Taxa contributions to community synchrony

We further estimated taxa-scale contributions to community synchrony by randomizing the abundance of a given taxa and estimating how this randomization influenced community synchrony (Pignatelli & Dallas, 2026). We performed the randomization 999 times per taxa and for each year of sampling, calculating z-scores as a measure of divergence of true synchrony compared to the distribution of randomized synchrony values. By randomizing a single taxa while maintaining the other taxa abundances, we are able to isolate the effect of randomizing a single taxa, allowing the estimation of the contribution of that taxa to overall community synchrony. Importantly, contributions to synchrony can be negative—though this is not routinely observed—if randomizing the abundance of a taxa actually increases community synchrony.

Further, we standardized taxa abundance to explore if taxa-scale contributions to synchrony could be driven solely by numerical dominance relative to the temporal structure of population dynamics. To do this, we performed min-max standardization, bounding the abundance time series between 0 and 1. This allowed an empirical test of the role of relative abundance on contributions to synchrony, as well as an exploration of what aspects of population dynamics made a given taxa a strong contributor to synchrony. We hypothesized that the temporal consistency of a taxa through time might relate to taxa contribution to synchrony, with less temporally variable taxa contributing more strongly to community synchrony. To test this, we estimated the coefficient of variation for each taxa and each year, relating these values to contributions to community synchrony with linear models.

3 | RESULTS

3.1 | Environmental variability and community synchrony

Marine zooplankton communities tended to be significantly synchronous (Figure 2c), suggesting that temporal fluctuations in the abundance across taxa tended to be related, driven potentially by similar responses to environmental variation or other potential drivers (Allen et al., 2008). Importantly, species–environment relationships across zooplankton taxa could still be quite different, but they only need to be similar in sign to produce enhanced synchrony, especially in a temporally autocorrelated environment. Of the 37 years sampled, only 10 had non-significant community synchrony relative to the null model randomizations (Figure 2). There was no clear temporal pattern in community synchrony, suggesting that this aggregate property of the community is fairly stable even given environmental changes and

FIGURE 2 Community synchrony—measured for each year of sampling—was unrelated to annual estimates of temperature variability (a) or salinity variability (b), where dashed lines indicate linear model fits. Further, community synchrony varied over time within a fairly small range of synchrony values (c), though some years were significantly different from a null and others were not. Point colour indicates year of sampling, with filled points having significantly positive synchrony relative to null model randomizations.

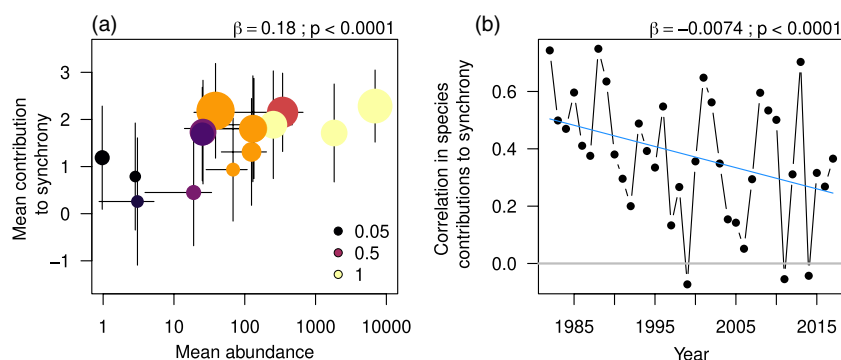
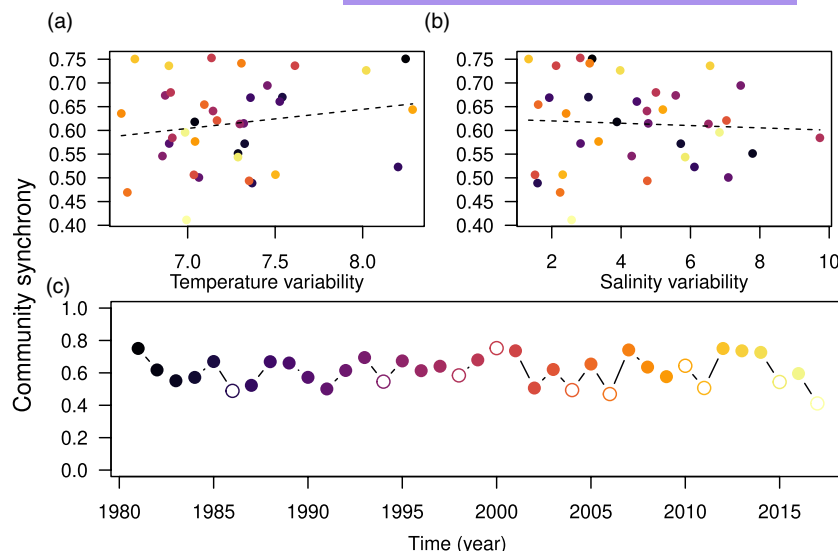


FIGURE 3 Contributions to synchrony (z-scores relative to randomized null communities) were positively related to taxon mean abundance, suggesting that more abundant taxa contribute more strongly and positively to synchrony (a; point size is proportional to the number of years the taxa contributed significantly to community synchrony, and point colour is proportional to temporal occupancy, quantified as the mean fraction of sampling events where the taxa was present). Further, correlations in these contributions over time (lag-1 correlations are plotted in b) suggest that the consistency of taxa contributions to synchrony have become weaker over time. Blue line is a linear model fit for visualization purposes.

variable population dynamics (see [Supporting Information](#)). Finally, we observed no clear relationship between temporal variability in the environment (temperature and salinity) and community synchrony ([Figure 2](#)).

3.2 | Taxa contributions to community synchrony

Taxa contributions to community synchrony varied through time ([Figure S1](#)), but more numerically dominant taxa tended to contribute more strongly to community synchrony. Further, taxon conservation of contribution to synchrony, estimated as the correlation in z-scores across taxa from 1 year to the next, varied over time. Taxa contributions to synchrony tended to be positively related when sampling first started in 1985, but have been tending towards zero in more recent years ([Figure 3b](#)). The importance of a given taxon to community synchrony was quantified both in terms of the number of years in which a significant positive contribution to synchrony was observed for a

given taxa ([Figure S1](#)) as well as in the mean contribution to community synchrony ([Figure 3a](#)), estimated as the departure from simulated null community synchrony when the target taxon abundance was randomized ($\beta = 0.18$, $p < 0.0001$, $R^2 = 0.41$).

Importantly, the relationship between contribution to community synchrony and mean abundance is not driven by mean abundance itself, but likely that more numerically dominant species were more stable (i.e. lower coefficient of variation in their abundance time series), as these relationships were insensitive to standardization of abundance. This highlights that it is not simply that the more numerically dominant taxa contribute more to synchrony (and less to overall community variability), but that more consistently present and stable taxa tend to contribute more strongly to community synchrony ([Figure 4](#)). That is, after scaling taxa abundance to have the same bounds, less temporally variable species still contributed more to community synchrony than more temporally variable taxa (unstandardized: $\beta = -0.23$, $p < 0.0001$, $R^2 = 0.11$; standardized: $\beta = -0.24$, $p < 0.0001$, $R^2 = 0.10$). Standardization of taxa abundance

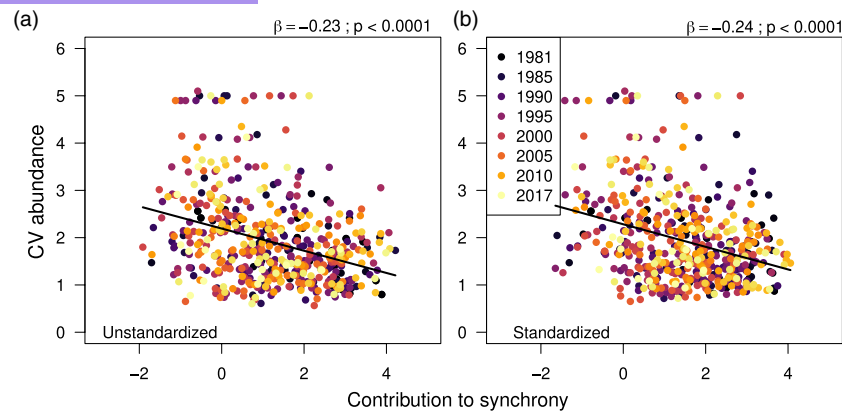


FIGURE 4 Contributions to synchrony (z-scores relative to randomized null communities) were negatively related to population variability (measured as the coefficient of variation), suggesting that taxa contributing most strongly to synchrony also tended to have less temporal variability in their population dynamics (i.e. they were more stable). This held true for both unstandardized (a) and standardized (b) abundance. Each point corresponds to a particular taxa for a given year, and point colour indicates year of sampling, with purple corresponding to older sampling years and yellow to more recent years.

did not strongly influence estimation of contributions to community synchrony (see [Figure S8](#)).

It is important to note that the coefficient of variation in abundance is sensitive to the temporal occupancy of a taxa in a given year. That is, taxa with many embedded zero abundance values, in which they were at too low of a density to be in a given sample, will tend to be more variable. We explore this in the [Supporting Information](#) and make it clear in [Figure 3](#) that temporal occupancy is related to mean abundance and contribution to synchrony. Future theoretical work is warranted to tease apart the role of temporal occupancy from variability in abundance, as well as to explore what happens when mean abundance is independent of variability in abundance.

4 | DISCUSSION

Throughout the 37-year time series, we observed consistent community synchrony for our set of marine zooplankton. However, this observed community synchrony was unrelated to environmental variability (temperature and salinity). The most consistent dominant zooplankton taxa were copepods, which was also the group contributing most strongly to community synchrony. Mean abundance was positively related to that taxa's contribution to community synchrony, and this was true even when taxon abundance was standardized prior to the estimation of community synchrony. This suggests that numerical dominance in itself is not driving contributions to community synchrony, but the relationship between numerical dominance and population variability (i.e. more abundant taxa tend to be less variable; Henderson & Magurran, 2014). We found that less temporally variable taxa had the highest contributions to community synchrony. Finally, the consistency of taxa contributions to synchrony through time has been decreasing through time (estimated as the year to year rank correlation in taxa contributions to synchrony), suggesting that the relative strength of taxa contributions may be

shifting as taxa respond differently to ecological pressures such as changing climatic drivers and species interactions.

Taxa with more stable dynamics contributed more strongly to community synchrony, providing a clear link between community synchrony and population variability. There is a demonstrated relationship between community synchrony and community stability (Blüthgen et al., 2016; Valencia et al., 2020; van Klink et al., 2019; Wagg et al., 2018) with more synchronous communities tending to be less stable, with stability typically measured as the coefficient of variation in overall community size (Donohue et al., 2016). By exploring species contributions to community synchrony, and linking these contributions to population dynamics, we provide a step towards understanding how the perturbation of a taxa or functional group might contribute to overall community synchrony. That is, historically stable species may contribute more to overall synchrony, even if they are not numerically dominant, and perturbations to these species will likely have a much stronger effect on community synchrony relative to more temporally variable species.

It is worth noting that taxa contributions to synchrony could be negative, suggesting that the randomization of that taxa's timeseries actually increased community synchrony. This does not mean that the wholesale removal of this taxa from the community would appreciably or predictably change community synchrony or stability, as communities are inherently dynamic, constrained by competitive interactions which may shift with species removal or additions. However, this is an interesting aspect of taxa, leaving clear open questions about the underlying demographic or trait basis for these species that essentially contribute negatively to community synchrony.

Despite the consistent role of taxon variability on community synchrony (e.g. more stable taxa contributed more positively to community synchrony), the temporal consistency of taxa's relative contributions to community synchrony has changed over time. That is, between any two consecutive years, we would expect that taxa that contributed strongly to synchrony in 1 year would

likely contribute strongly in the next. While this was true in the 1980s in our system, the correlation in taxa contributions to community synchrony over consecutive timesteps has become far less positive. There are many potential drivers of relative contributions of different taxa to overall community synchrony (Pignatelli & Dallas, 2026), such as evolutionary processes, changes in taxa phenology, abundance of natural enemies or a number of other processes. It is also unclear if this decline in consistency of contribution to synchrony actually has potential to influence overall community synchrony, as this decline in temporal consistency of relative contributions to community synchrony did not have an effect on community synchrony itself. This suggests potentially that taxa may change in their relative contributions through time while serving functionally redundant roles to overall community synchrony.

The lack of importance of environmental variability on community synchrony is counter to how we think communities respond to fluctuating environments (Chesson & Huntly, 1997; Gonzalez & Descamps-Julien, 2004; Loreau & de Mazancourt, 2008; Tuljapurkar, 2013). It is possible that temperature and salinity are not strong drivers of marine zooplankton density in this system (Allen et al., 2008), or potentially that species had similar responses to environmental drivers that overall community synchrony was not strongly influenced (e.g. taxa demographic rates change with environment, but in a way that does change community synchrony, as community synchrony and stability are measures of variability in response, a coarser measure than demographic rate change). Importantly, this estuarine community may be driven primarily by the environmental conditions and species dynamics in the open ocean (Buzzelli et al., 2009). Residence time for the sampled location is approximately 15 h, highlighting the dynamic nature of this community. Further, we do not consider the role of trophic interactions among taxa, which could drive population dynamics (Verhoef & Olff, 2010). A more resolved food web and finer-scale taxonomic identification could alleviate this current constraint. Finally, the identification of taxa to functional groups or higher level taxa must be recognized as a factor that could influence our findings. The logistics of identifying these taxa to species is daunting, and in some cases infeasible. Numerically dominant groups such as copepods contain numerous species (at least 10 likely Mallick et al. (2024)), all of which could hypothetically respond differently to environmental variation. This would create a situation where taxa appear to be quite temporally stable and contribute strongly to community synchrony, but they do so only as a composite, where individual species could be quite variable. This creates what is essentially a taxa-level portfolio effect, in which species fluctuations can be quite large, but the aggregate taxa-level density may remain quite stable. Finally, the role of seasonal dynamics and the importance of time-scale dependence should be recognized. Sampling was performed every 2 weeks, but the generation time of many of these taxa is shorter. Seasonal changes in abundance (see Figure S2) may also influence community synchrony (Shoemaker et al., 2022).

Marine zooplankton communities were consistently synchronous over the 37-year time series, even when the contribution of each taxa to community synchrony varied through time. We tested whether this variation in contributions to synchrony was being driven by numerically dominant taxa by comparing results of standardized and unstandardized abundance. We found that it is not necessarily the overall numerical dominance that makes taxa strong contributors to community synchrony, but the structure of their fluctuations in abundance (i.e. their temporal variability or *stability*). This highlights that the temporal structure of species population dynamics may be more important than overall numerical dominance, providing a clear link between synchrony and stability that many have explored at the community-level previously (Blüthgen et al., 2016; Valencia et al., 2020; van Klink et al., 2019; Wagg et al., 2018). Future efforts to link species contributions to community synchrony to theoretical models of community dynamics will provide more concrete connections between classic community assembly theory (May, 1972, 1973) and explorations of community synchrony (Loreau & de Mazancourt, 2008; Shoemaker et al., 2022; Valencia et al., 2020).

AUTHOR CONTRIBUTIONS

Joshua Stone collected the data and helped design the project. Lauren Holian and Anthony Pignatelli helped design the analysis and contributed to manuscript writing. Tad A. Dallas analysed the data and wrote the initial draft of the manuscript. All authors contributed to manuscript editing.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest to declare.

DATA AVAILABILITY STATEMENT

Data and R code are available on figshare at <https://doi.org/10.6084/m9.figshare.28832429> (Dallas et al., 2026).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Figure S1. Taxa contributions to community synchrony were fairly consistent over time, though more abundant taxa (point size is proportional to log-transformed abundance) tended to have more variable contributions. Taxa are ordered by their average abundance, with copepods being the most abundant sampled taxa. The legend provides the number of sampled years where that taxa contributed significantly positively to synchrony. Horizontal lines at 1.96 and -1.96 mark significance thresholds at $\alpha = 0.05$.

Figure S2. Zooplankton densities broken down by the composite taxa over time (purple corresponds to earlier years in sampling and yellow to later years of the time series spanning 1981–2017).

Figure S3. The relationship between standard deviation (SD) and proportional variability (PV) for temperature (left panel) and salinity (right panel), where each point corresponds to a yearly estimate of environmental variability.

Figure S4. Contributions to synchrony (z-scores relative to randomized null communities) were positively related to abundance, suggesting that taxa contributing most strongly to synchrony also tended to have higher abundance. This held true for both unstandardized (left) and standardized (right) abundance. Each point corresponds to a particular taxa for a given year, and point colour indicates year of sampling, with purple corresponding to older sampling years and yellow to more recent years.

Figure S5. Temporal occupancy was positively related to species contribution to synchrony, though more weakly than the relationship between taxa variability (CV in abundance). This held true for both unstandardized (left) and standardized (right) abundance. Each point corresponds to a particular taxa for a given year, and point colour

indicates year of sampling, with purple corresponding to older sampling years and yellow to more recent years.

Figure S6. Temporal occupancy was negative related to temporal variability in taxa abundance, suggesting that taxa with many embedded zero values (i.e. those with lower temporal occupancy) also tended to have more variable abundance time series. Each point corresponds to a particular taxa for a given year, and point colour indicates year of sampling, with purple corresponding to older sampling years and yellow to more recent years. Plotted line is a negative exponential model fit.

Figure S7. Taxa contributions to community synchrony were fairly consistent over time, though more abundant taxa tended to have more variable contributions. Taxa are ordered by their average abundance, with copepods being the most abundant sampled taxa. The legend provides the number of sampled years where that taxa contributed significantly positively to synchrony. Horizontal lines at 1.96 and -1.96 mark significance thresholds at $\alpha = 0.05$.

Figure S8. Standardization of taxa density did not influence the estimation of taxa contributions to community synchrony, as we observed a strong positive relationship between unstandardized (y-axis) and standardized (x-axis) contributions to synchrony.

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